

New Feature Selection Pipeline

LOOCV Benchmark of 28 FS Methods on Tapping Features

Gary Wang

Nordling Lab @ Dept. of Mechanical Engineering
National Cheng Kung University





Motivation

- Dataset: **n=53** samples, **107** numeric tapping features.
- Target: MDS-UPDRS finger-tapping rating (0–4 ordinal score).
- Goal: identify which feature-selection (FS) method gives the most predictive, compact subset.
- Approach: hold the predictor fixed, vary the FS method → FS effect is isolated.





Pipeline Overview

Stage 1: Run **28 FS methods** (filter / embedded / wrapper) on 107 features



Stage 2: For each subset → **LOOCV** with fixed predictor (MLP-8, tanh, L2)



Stage 3: Score with **4 metrics** (MAE, Spearman, Kappa, Adj. Acc.) + 2000-iter bootstrap CIs



Stage 4: Rank FS methods by **Composite** = mean of normalised metrics



Stage 5: Among well-performing FS methods, count **feature inclusion frequency**





Feature Groups Overview — Unified Clinical Grouping (107)

Group	Clinical Domain	N
A. Kinematics	Raw finger displacement range & variability	24
B. Speed	Tapping rate, opening/closing velocity, and their decrement	23
C. Amplitude	Finger-opening size, shape, and progressive decrement	25
D. Rhythm	Temporal regularity of inter-tap intervals	14
E. Freezing	Movement pauses and halts	7
F. Task Completion	Ability to perform the full 10-tap sequence	5
G. Phase Asymmetry	Timing asymmetry between opening and closing	9
Total		107

*Features from v1 and v2 are merged into clinically motivated groups.
Each group captures a distinct aspect of the MDS-UPDRS 3.4 finger-tapping assessment.*



Signal Preprocessing

Raw input: 2D coordinates of thumb & index finger per frame.

Distance	$\sqrt{(\Delta X)^2 + (\Delta Y)^2}$ between fingers
Angle	$\frac{\text{Distance}}{\max(\text{Distance})} \times 90^\circ$
Speed	$d(\text{Angle})/dt$
Accel.	$d(\text{Speed})/dt$

Peak = frame of maximum finger separation (tap).

Valley = frame of minimum separation between taps.

Per-tap amplitude:

$$A_k = d(\text{peak}_k) - \frac{d(v_{\text{before}}) + d(v_{\text{after}})}{2}$$

Inter-tap interval (ITI):

$$\text{ITI}_k = t(\text{peak}_{k+1}) - t(\text{peak}_k)$$

Opening phase of tap k : $v_{\text{before}} \rightarrow \text{peak}_k$

Closing phase of tap k : $\text{peak}_k \rightarrow v_{\text{after}}$





Group A: Kinematics (24)



Feature	Definition
finger_mvmnt_x_*(6)	$ \Delta X_{\text{thumb}} $, 6 stats
finger_mvmnt_y_*(6)	$ \Delta Y_{\text{thumb}} $, 6 stats
finger_mvmnt_dist_*(6)	$\sqrt{\Delta X^2 + \Delta Y^2}$, 6 stats
acceleration_*(6)	$ d(\text{Speed})/dt $, 6 stats

Stats: median, IQR, mean, min, max, std

Motivation

PD patients exhibit reduced range of motion and increased spatial variability due to bradykinesia and rigidity.

Displacement features capture the *spatial scale* of finger movement.

Acceleration captures the *sharpness* of tap initiation and termination — sluggish starts and stops are characteristic of PD but not always visible in mean speed.



Group B: Speed (23)

Feature	Definition
speed_*(6)	$ d(\text{Angle})/dt $, 6 stats
frequency_*(6)	$f_k = 1/ IT _k$ (Hz), 6 stats
freq_lr_slope	Slope of linear fit to f_k ; negative = slowing
freq_lr_fitness_r2	R^2 of the frequency-time fit
freq_fit_min_degree	Fixed = 1 (linear model)
V_open_*(3)	$\max(\dot{d})$ during opening, 3 stats
V_close_*(3)	$ \min(\dot{d}) $ during closing, 3 stats
R_V_open	$\text{med}(V_{\text{open}}[-3 : -1]) / \text{med}(V_{\text{open}}[1 : 3])$
R_V_close	Same ratio for closing velocity

Motivation

Tapping speed directly reflects bradykinesia. Progressive deceleration ($\text{freq_lr_slope} < 0$) is a key MDS-UPDRS 3.4 sub-criterion.

R^2 distinguishes clean monotone slowing from erratic fluctuation.

Per-phase velocity ($V_{\text{open/close}}$) captures asymmetric impairment between opening and closing that whole-tap speed misses.

$R_{V_{\text{open/close}}}$ measure velocity decrement analogous to amplitude decrement.





Group C: Amplitude (25)

$$A_k = d(\text{peak}_k) - \text{mean}(\text{adjacent valleys})$$

Feature	Definition
amplitude_*(6)	6 stats of $\{A_k\}$
amplitude_cv	σ/μ of $\{A_k\}$
amplitude_entropy	Entropy of A_k histogram
amplitude_MAD/skew/kurtosis	Robust spread, asymmetry, tail weight
amp_decr_slope	Slope of linear fit to A_k
amp_decr_fitness_r2	R^2 of the linear fit
amp_decr_end_to_mean	$A_{\text{last}}/\bar{A}$
amp_decr_last_to_first_half	$\bar{A}_{\text{2nd half}}/\bar{A}_{\text{1st half}}$
A_early, A_late	Median of first / last 3 taps
R_A_3_3	$A_{\text{late}}/A_{\text{early}}$
D_A_pct	$100(1 - R_{A_3_3})$
D_A_max	$1 - A_{\text{min}}/A_{\text{max}}$
k_collapse	First k s.t. $A_k <$

Motivation

Reduced amplitude is the most salient MDS-UPDRS 3.4 criterion.

Basic stats + CV + entropy capture overall scale and tap-to-tap inconsistency. MAD/skew/kurtosis reveal outlier taps that std masks.

Linear decrement features quantify progressive amplitude reduction from multiple angles (trend, endpoint, half-to-half).

The supervisor's formula uses A_{early} as a patient-specific baseline, making $R_{A_3_3}$ and k_{collapse} independent of absolute amplitude.





Group D: Rhythm (14)

Feature	Definition
period_*(6)	$ITI_k = t(\text{peak}_{k+1}) - t(\text{peak}_k)$ in s, 6 stats
aperiodicity	Shannon entropy of FFT power spectrum
periodEntropy	Entropy of ITI histogram (50 bins, 0–2 s)
periodVarianceNorm	$\text{var}(ITI) / \max(ITI)$
I_ITI	$\text{mean}(ITI_{k+1} - ITI_k)$ (MASD)
I_ITI_norm	$I_{ITI} / \bar{ITI}$
ITI_MAD	Median absolute deviation of ITI
ITI_skew	Skewness of ITI distribution
ITI_kurtosis	Kurtosis of ITI distribution

Motivation

Period stats give mean rhythm speed (std/IQR quantify stability).

aperiodicity and periodEntropy capture global irregularity from two independent perspectives (frequency domain vs. time domain).

I_{ITI} (analogous to MSSD) measures short-term instability between adjacent taps — more sensitive to sudden timing errors than global std.

MAD, skewness, and kurtosis of ITI further characterise the shape of the timing distribution.





Groups E & F: Freezing (7) & Task Completion (5)

E — Freezing / Interruption

Feature	Definition
numInterruptions	Count of low-speed events $>$ interruption threshold
numFreeze	Count of low-speed events $>$ freeze threshold
maxFreezeDuration	Duration (s) of longest freeze event
num_inter_norm	$\text{numInterruptions} / \text{num_peaks}$
num_freeze_norm	$\text{numFreeze} / \text{num_peaks}$
S_halt	Total cumulative halt time (s)
F_halt	$S_{\text{halt}} / \text{total recording duration}$

Motivation

Motor freezing and hesitation are direct MDS-UPDRS indicators. Rate-normalised counts (`_norm`) decouple pause frequency from tapping speed; `S_halt` and `F_halt` capture cumulative time lost to motor blocks.

MDS-UPDRS 3.4 asks for 10 taps. Inability to complete is itself a severity indicator. `F_incomplete` provides a continuous version of the binary completion criterion, comparable across patients.

F — Task Completion

Feature	Definition
num_peaks / N_valid	Total valid taps detected
C_10	1 if $N_{\text{valid}} \geq 10$, else 0
T_10	Time (s) from tap 1 to tap 10
F_incomplete	$\max(0, 1 - N_{\text{valid}} / 10)$





Group G: Phase Asymmetry (9)

Each tap split into Opening ($v_{\text{before}} \rightarrow \text{peak}$) and Closing ($\text{peak} \rightarrow v_{\text{after}}$).

Feature	Definition
$T_{\text{open_}}^*(3)$	Opening phase duration (s), 3 stats
$T_{\text{close_}}^*(3)$	Closing phase duration (s), 3 stats
$R_{\text{OC_}}^*(3)$	$T_{\text{open}}/T_{\text{close}}$ per tap, 3 stats

Note: opening/closing velocity ($V_{\text{open/close}}$) is in Group B.

Motivation

PD patients often show asymmetric impairment between opening (finger separation) and closing (finger approximation).

$R_{\text{OC}} < 1$ indicates closing takes disproportionately longer, associated clinically with difficulty in muscle relaxation.

Separating the two phases captures information that disappears when only full amplitude or full period is measured.





28 Feature Selection Methods



Filter (12)

- Pearson
- Spearman
- Kendall
- Mutual Info
- F-regression
- mRMR
- Distance Correlation
- HSIC
- Fisher Score
- ReliefF
- FCBF
- Variance Threshold

Embedded (11)

- LASSO (Standard)
- ElasticNet
- Adaptive LASSO
- LARS
- RF Importance
- Extra Trees
- Gradient Boosting
- XGBoost
- Permutation Importance
- Stability Selection
- Boruta

Wrapper (5)

- RFECV (RF)
- RFECV (SVR)
- RFECV (Lasso)
- SFS Forward
- SBS Backward



Evaluation Protocol

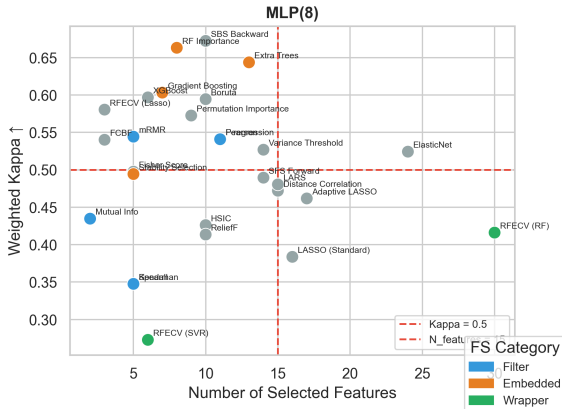
- **Predictor:** MLPRegressor(hidden=(8,), tanh, lbfgs, $\alpha=1.0$)
- **Validation:** Leave-One-Out CV (53 folds).





Compactness vs. Performance

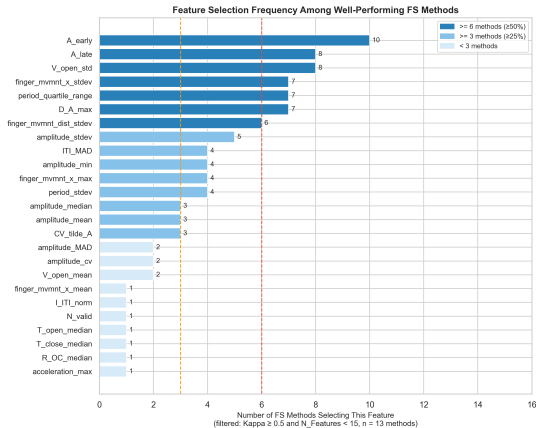
Number of Features vs Weighted Kappa
(red lines = Stage 5 filter: $Kappa \geq 0.5$ AND $N_{features} < 15$)



Red lines = Stage 5 filter ($Kappa \geq 0.5$, $N_{feat} < 15$); 17/28 methods pass.



Feature Inclusion Frequency



Filter: only FS methods with Kappa ≥ 0.5 AND $N_{\text{feat}} < 15$ contribute (17 methods). Features picked by $\geq 50\%$ of these methods are strong candidates.



Take-aways

- **Embedded & wrapper** methods dominate; pure-correlation filters underperform.
- **Adaptive LASSO, LASSO, LARS** achieve highest Kappa (≈ 0.74) with only 8 features.
- **SBS Backward** gives best Spearman (0.783) and MAE (0.302) – most balanced.
- **RFECV (SVR)** is most compact: competitive Kappa (0.72) with only 6 features.
- 17/28 FS methods pass the Stage 5 filter; their frequency analysis gives a principled feature short-list for the LLM-Lasso prompt.
- Outputs in `fs_results_v2/{png,pdf}/`.



Thank you!

Questions & Discussion

